Question 1) Probability density and current: The time-dependent Schrödinger equation in 3D is

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x},t) = -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{x},t) + V(\mathbf{x})\psi(\mathbf{x},t).$$

Using the Schrödinger equation, we can calculate the probability density $\rho(\mathbf{x},t) = |\psi(\mathbf{x},t)|^2$, which gives the probability of finding the particle at an infinitesimally small region around $\mathbf{x}$ at time t. For a closed system, the probability density should satisfy the continuity equation:

$$\frac{\partial\rho(\mathbf{x},t)}{\partial t} + \nabla\cdot\mathbf{j}(\mathbf{x},t) = 0. \tag{1}$$

This is a statement of local conservation of probability, with $\mathbf{j}(\mathbf{x},t)$ the probability current density (how much probability is flowing through unit area per unit time at given $\mathbf{x}$ and t).

a) Derive the standard expression for the probability current density:

$$\mathbf{j}(\mathbf{x},t) = \frac{\hbar}{2mi}\left(\psi^*(\mathbf{x},t)\nabla\psi(\mathbf{x},t) - \psi(\mathbf{x},t)\nabla\psi^*(\mathbf{x},t)\right).$$

Use the continuity equation in Eq. (1).

b) Now consider a polar form for the wave function: $\psi(\mathbf{x},t) = \sqrt{\rho(\mathbf{x},t)}e^{iS(\mathbf{x},t)/\hbar}$, where $\rho(\mathbf{x},t)$ is the probability density and $S(\mathbf{x},t)$ is the phase of the wave function. Show that the probability current density can be written as

$$\mathbf{j}(\mathbf{x},t) = \frac{\rho(\mathbf{x},t)}{m}\nabla S(\mathbf{x},t).$$

What does this mean for a *real* wavefunction $\psi(\mathbf{x})$?

c) For a plane-wave solution of the form $\psi(\mathbf{x},t) = \frac{1}{\sqrt{V}}e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$, where V is some normalization volume (formally, $V\to\infty$ for a bona-fide plane wave), $\mathbf{k}$ is the wave vector, and ω is the angular frequency, compute the probability density $\rho(\mathbf{x},t)$. Is it time-dependent? How about the probability current density $\mathbf{j}(\mathbf{x},t)$?

Question 2) The Propagator: The propagator $K(x_1,t_1;x_0,t_0)$ is defined as the transition amplitude between two space-time points, obtained by applying the time-evolution operator $U(t_1,t_0)$ between position eigenstates:

$$K(x_1,t_1;x_0,t_0) = \langle x_1|\,\hat{U}(t_1,t_0)\,|x_0\rangle = \langle x_1|\,e^{-\frac{i}{\hbar}\hat{H}(t_1-t_0)}\,|x_0\rangle\,, \tag{2}$$

with the latter equation valid for a time-independent Hamiltonian $\hat{H}$. We will prove a few properties of the propagator in this exercise.

a) Show that the propagator obeys Schrödinger's equation by differentiating the definition of the propagator with respect to t_1 and using $\partial_{t_1}U(t_1,t_0) = -\frac{i}{\hbar}\hat{H}(t_1)U(t_1,t_0)$.

b) Show that the wavefunction at time t_1 can be expressed in terms of the propagator and the initial wavefunction at time t_0 as

$$\psi(x_1,t_1) = \int dx_0\, K(x_1,t_1;x_0,t_0)\psi(x_0,t_0), \tag{3}$$

where $\psi(x,t) = \langle x|\psi(t)\rangle$.

c) prove the following properties of the propagator:

i) **Initial condition**: At equal times, the propagator reduces to the Dirac delta function

$$K(x_1, t; x_0, t) = \delta(x_1 - x_0). \tag{4}$$

ii) **Composition**: The propagator satisfies the composition property

$$K(x_2, t_2; x_0, t_0) = \int dx_1 \, K(x_2, t_2; x_1, t_1) K(x_1, t_1; x_0, t_0), \tag{5}$$

for any intermediate time t_1.

Question 1) This exercise is a straightforward application of Schrödinger's equation.
a) Compute the time derivative of the probability density:

$$\frac{\partial \rho}{\partial t} = \frac{\partial}{\partial t}|\psi|^2 = \frac{\partial}{\partial t}(\psi^*\psi) = \psi^*\frac{\partial \psi}{\partial t} + \psi\frac{\partial \psi^*}{\partial t}. \tag{6}$$

We will need Schrödinger's equation and its complex conjugate:

$$\frac{\partial \psi}{\partial t} = \frac{1}{i\hbar}\left(-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi\right), \tag{7}$$

$$\frac{\partial \psi^*}{\partial t} = -\frac{1}{i\hbar}\left(-\frac{\hbar^2}{2m}\nabla^2\psi^* + V\psi^*\right). \tag{8}$$

Substituting these into the equation above,

$$\frac{\partial \rho}{\partial t} = \psi^*\left(\frac{1}{i\hbar}\left(-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi\right)\right) + \psi\left(-\frac{1}{i\hbar}\left(-\frac{\hbar^2}{2m}\nabla^2\psi^* + V\psi^*\right)\right) \tag{9}$$

$$= -\frac{\hbar}{2mi}\left(\psi^*\nabla^2\psi - \psi\nabla^2\psi^*\right) = -\nabla\cdot\left(\frac{\hbar}{2mi}\left(\psi^*\nabla\psi - \psi\nabla\psi^*\right)\right), \tag{10}$$

proving our result that

$$\mathbf{j}(\mathbf{x},t) = \frac{\hbar}{2mi}\left(\psi^*(\mathbf{x},t)\nabla\psi(\mathbf{x},t) - \psi(\mathbf{x},t)\nabla\psi^*(\mathbf{x},t)\right).$$

b) You can show this by direct substitution. Sparing us both from the details, I will write the first and last step:

$$\mathbf{j}(\mathbf{x},t) = \frac{\hbar}{2mi}\left(\sqrt{\rho}e^{-iS/\hbar}\nabla(\sqrt{\rho}e^{iS/\hbar}) - \sqrt{\rho}e^{iS/\hbar}\nabla(\sqrt{\rho}e^{-iS/\hbar})\right) = \frac{\rho}{m}\nabla S. \tag{11}$$

Note that we used the fact that $\nabla\rho = 2\sqrt{\rho}(\nabla\sqrt{\rho})$ and that $\nabla e^{iS/\hbar} = (i/\hbar)e^{iS/\hbar}\nabla S$.

Note that for a real wavefunction, the phase S is zero, so $\mathbf{j}(\mathbf{x},t) = 0$ (we mean $\psi(\mathbf{x})$ is real for all $\mathbf{x}$ and we are assuming a stationary state).



This result is quite interesting: it shows that the flow of probability depends on the rate of change of the phase of the wave function. With an abuse of notation, we may even interpret $\nabla S/m$ as a velocity field for the probability flow, $j = \rho\mathbf{v}$.

c) For the plane wave, the probability density is just a constant: $\rho(\mathbf{x},t) = |A|^2$, which is independent of time, $\partial\rho/\partial t = 0$. The LHS of the continuity equation is zero, so the RHS must also be zero: $\nabla\cdot\mathbf{j} = 0$. But what is $\mathbf{j}$? Using the result above, we have that $\nabla S = \hbar\mathbf{k}$, so the probability current density is

$$\mathbf{j}(\mathbf{x},t) = \frac{|A|^2}{m}\hbar\mathbf{k}.$$

Despite the fact that the probability density is constant in space and time, the probability density is constant flowing in the direction of the wave vector $\mathbf{k}$. This is consistent with our intuition of a plane wave representing a free particle with momentum $\mathbf{p} = \hbar\mathbf{k}$.

Question 2)

a) We start from the definition of the propagator:

$$K(x_1,t_1;x_0,t_0) = \langle x_1|\,\hat{U}(t_1,t_0)\,|x_0\rangle\,. \tag{12}$$

Differentiating with respect to t_1, we have

$$\frac{\partial}{\partial t_1}K(x_1,t_1;x_0,t_0) = \langle x_1|\,\frac{\partial\hat{U}(t_1,t_0)}{\partial t_1}\,|x_0\rangle\,. \tag{13}$$

Using the relation $\partial U(t_1,t_0)/\partial t_1 = -\frac{i}{\hbar}\hat{H}(t_1)U(t_1,t_0)$, we get

$$\frac{\partial}{\partial t_1}K(x_1,t_1;x_0,t_0) = -\frac{i}{\hbar}\,\langle x_1|\,\hat{H}(t_1)\hat{U}(t_1,t_0)\,|x_0\rangle \tag{14}$$

$$= \frac{-i}{\hbar}H(x,t_1)\,\langle x_1|\,\hat{U}(t_1,t_0)\,|x_0\rangle \tag{15}$$

$$= \frac{-i}{\hbar}H(x,t_1)K(x_1,t_1;x_0,t_0). \tag{16}$$

which is Schrödinger's equation for the propagator.



Let me pause to clarify the step in Eq. (15). It is simply the statement that for a general ket $|\psi\rangle$:

$$\langle x|\,\hat{H}(t_1)\,|\psi\rangle = \langle x|\,\Big(\hat{H}(t_1)\,|\psi\rangle\Big) = H(x,t_1)\psi(x,t_1) = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t_1)\right)\psi(x,t_1), \tag{17}$$

where we interpreted $\hat{H}(t_1)\,|\psi\rangle$ as the new vector that we are going to project onto the position basis using $\langle x|$. In that sense, it is not that we are passing $\hat{H}(t_1)$ through $\langle x_1|$, but rather we are applying the Hamiltonian operator to the ket first, and then projecting onto position, which gives us the differential operator acting on the wavefunction. And to clarify, the operator $\hat{H}(t_1)$ acts on the Hilbert space of kets, and the "operator" ∇^2, for example, actts on the space of functions of position.
In other words, the equation:

$$\langle x_1|\,\hat{H}(t_1)\,|\psi\rangle = H(x_1,t_1)\psi(x_1,t_1), \tag{18}$$

is telling you how to turn the operation on the Hilbert space into the operation of $H(x_1,t_1)$ on the space of functions of position. In the same way, the equation:

$$\langle x_1|\,\hat{U}(t_1,t_0)\,|x_0\rangle = K(x_1,t_1;x_0,t_0), \tag{19}$$

tells you how to turn the operation on the Hilbert space into the operation of $K(x_1,t_1;x_0,t_0)$ on the space of functions of position (in this case, a trivial multiplication by a complex exponential function).

b) The wavefunction at time t_1 is given by

$$\psi(x_1,t_1) = \langle x_1|\,\hat{U}(t_1,t_0)\,|\psi(t_0)\rangle\,. \tag{20}$$

Inserting a complete set of position eigenstates at time t_0, we have

$$\psi(x_1,t_1) = \int dx_0\,\langle x_1|\,\hat{U}(t_1,t_0)\,|x_0\rangle\,\langle x_0|\psi(t_0)\rangle\,. \tag{21}$$

Recognizing the first term as the propagator and the second term as the initial wavefunction, we obtain

$$\psi(x_1, t_1) = \int dx_0 K(x_1, t_1; x_0, t_0)\psi(x_0, t_0). \tag{22}$$

c) Initial condition: At equal times, the time-evolution operator reduces to the identity operator, $\hat{U}(t, t) = \mathbb{1}$. Thus, the propagator becomes

$$K(x_1, t; x_0, t) = \langle x_1 | \, \mathbb{1} \, | x_0 \rangle = \langle x_1 | x_0 \rangle = \delta(x_1 - x_0). \tag{23}$$

Composition: We start from the definition of the propagator:

$$K(x_2, t_2; x_0, t_0) = \langle x_2 | \, \hat{U}(t_2, t_0) \, | x_0 \rangle = \langle x_2 | \, \hat{U}(t_2, t_1)\hat{U}(t_1, t_0) \, | x_0 \rangle \, , \tag{24}$$

where we are using the composition property of the time-evolution operator (this one is trivial to prove as well). Inserting a complete set of position eigenstates at an intermediate time t_1, we have

$$K(x_2, t_2; x_0, t_0) = \int dx_1 \, \langle x_2 | \, \hat{U}(t_2, t_1) \, | x_1 \rangle \, \langle x_1 | \, \hat{U}(t_1, t_0) \, | x_0 \rangle \, . \tag{25}$$

Recognizing the terms as propagators, we obtain

$$K(x_2, t_2; x_0, t_0) = \int dx_1 K(x_2, t_2; x_1, t_1)K(x_1, t_1; x_0, t_0). \tag{26}$$